# Lifeng Han, PhD

Computational Linguistics | Multilingual NLP | Language-Model Evaluation

l.han@liacs.leidenuniv.nl
ORCID: 0000-0002-3221-2185 | Google Scholar | Leiden profile

## Academic Profile

Computational linguist investigating linguistically grounded and human-centred evaluation of multilingual language technologies. Research spans machine translation metrics (LEPOR), Chinese linguistic representations, multi-word expressions (MultiMWE and AlphaMWE), LLM translation blind spots, metaphor interpretation (MetaHOPE and NewsMeta), and reliable NLP in healthcare. Research leadership includes independently designing methods and software, coordinating multilingual resources and collaborative publications, and supervising student-led research. University-level teaching and undergraduate/master's supervision complement an interdisciplinary research agenda connecting linguistics, AI, and cognitive science.

## Research Areas

Computational linguistics; cross-linguistic and multilingual NLP; lexical semantics and multi-word expressions; machine translation and evaluation; metaphor and figurative language; LLM evaluation and reliability; language resources; human-centred AI; clinical NLP and retrieval-augmented generation.

## Academic Appointments

**Postdoctoral Research Associate (80%), Leiden University Medical Center, Netherlands** Oct 2024–Sep 2026
AI for Health; EU 4D PICTURE project.

**Lecturer (20%), Leiden Institute of Advanced Computer Science, Leiden University** Nov 2024–Oct 2026
Master's-level teaching and student research supervision.

**Postdoctoral Research Associate, University of Manchester, UK** Mar 2022–Sep 2024
NLP Group; UKRI/EPSRC HealthcareNLP programme.

**Part-time Lecturer, Manchester Metropolitan University, UK** Feb–Mar 2023
MSc AI and Machine Learning: Business Applications (five weeks of a ten-week module).

**Research Assistant, ADAPT Research Centre, Dublin City University** Oct–Nov 2016

**Doctoral Researcher / Guest Researcher, ILLC, University of Amsterdam** Sep 2014–Sep 2016
Machine translation, paraphrasing and evaluation.

**Research Assistant, NLP2CT Laboratory, University of Macau** Oct 2011–Aug 2014
Chinese–Portuguese translation, syntax, evaluation, and multilingual resources.

## Education

**PhD, Computer Applications (Natural Language Processing), Dublin City University** Awarded 2022
ADAPT Research Centre; research on multi-word expressions in machine translation. Doctoral research 2016–2021.

**MSc, Software Engineering, University of Macau** 2014
Excellent Award.

**BSc, Mathematics and Applied Mathematics, Hebei Normal University** 2011

## Selected Original Contributions and Research Leadership

- **LEPOR (COLING 2012):** personally designed and coded the machine translation evaluation metric; continued metric research through cushLEPOR (WMT 2021).
- **Chinese decomposition for MT (ESSLLI 2018; NoDaLiDa 2021):** designed and experimentally evaluated radical- and stroke-level representations for low-frequency Chinese lexical items.
- **MultiMWE (LREC 2020):** created the automatic multilingual MWE parallel-resource construction approach.
- **AlphaMWE (COLING 2020; LRE 2026):** coordinated development of multilingual parallel corpora annotated for multi-word expressions.
- **Mind the Gap (WMT 2026):** coordinated collaborative investigation of LLM translation blind spots using AlphaMWE.
- **MetaHOPE (ICNLSP 2026) and NewsMeta (submitted to LRE):** supervised research on metaphor-oriented translation evaluation and bilingual metaphor-annotated news data.
- **Clinical NMT (Frontiers in Digital Health 2024):** research coordination and scientific advisory contributions in collaboration with US language-technology company Logrus Global.

## Teaching and Student Supervision

- **Leiden University (2024–2026):** lecturer, master's course in Business Research Methods; supervised/co-supervised BSc and MSc theses on NLP, language models, topic modelling, knowledge graphs and clinical text.
- **Manchester Metropolitan University (2023):** delivered five weeks of master's-level AI and Machine Learning: Business Applications.
- **University of Manchester (2022–2024):** supervised/co-supervised student research in clinical NLP, temporal relations, machine translation, synthetic clinical text, information extraction and biomedical readability. Student projects produced publications at ACL SRW, IEEE ICHI/BigData, EAMT/AMTA and journals.
- **Earlier teaching:** co-lecturer in research methods, mathematics and introductory programming at Pulse College (2021–2022); co-designed an approved mathematics-for-game-development module; Operating Systems co-tutor at DCU (2020); Python teaching assistant at University of Amsterdam (2014).
- **Mentoring approach:** research question formulation, linguistic annotation, reproducible experiments, evaluation and scientific writing; multiple supervised BSc/MSc projects led to peer-reviewed papers.

## Research Funding and Funded Projects

- **Awarded:** two University of Manchester Open Research Fund awards (2022; GBP 2,000 each) for clinical NLP projects; funded Science and Engineering Summer Research Internship (2023–2024; 12-week intern stipend of GBP 1,500/month plus GBP 500 research costs).
- **Project participation:** EU 4D PICTURE, via Leiden University Medical Center (2024–2026); UKRI/EPSRC HealthcareNLP (EP/V047949/1), University of Manchester (2022–2024); ADAPT research programme, Dublin City University (2016–2021).
- **Competitive proposals:** JSALT summer workshop collaborative proposal ranked third in final round (2022–2023); Irish Research Council postdoctoral proposal scored 86.5/100 (2022). Neither is listed as awarded funding.

## Academic Leadership and Service

- Program Co-Chair, MWE 2023 (including ClinicalNLP track); Program Co-Chair, joint MWE–UD 2024 workshop.
- Standing Committee, ACL SIGLEX-MWE (2022–2024); area-chair/meta-review responsibilities at ACL 2023 and EMNLP 2023.
- Local organiser and tutorial coordination, HealTAC 2023; LREC 2022 tutorial on MT meta-evaluation; research seminars and conference tutorials.
- Member, ISO/NSAI committee activities related to AI and NLP evaluation (2020–2022); reviewer for computational linguistics conferences and journals.

## Selected Publications: Computational Linguistics and Multilingual NLP

*Fifteen representative peer-reviewed or accepted works selected for relevance to linguistic structure, cross-linguistic NLP, lexical and figurative semantics, language resources, and translation evaluation. All entries are also included in the complete publication list below.*

1. L. Han et al.. "Mind the Gap: Exposing LLM Translation Blind Spots Using the AlphaMWE Multilingual Parallel Corpus." *WMT 2026*.
2. J. Liang and L. Han. "MetaHOPE: A Metaphor-Oriented Evaluation Framework for Analysing MT and LLM Translation Errors." *ICNLSP 2026*.
3. J. Liang and L. Han. "Towards End-to-End Multilingual Metaphor Processing: Integrating Detection, Translation, and Evaluation." *AACL-IJCNLP 2026 Student Research Workshop (accepted)*.
4. L. Han et al.. "Towards a Resource for Multilingual Lexicons: An MT Assisted and Human-in-the-Loop Multilingual Parallel Corpus with Multi-Word Expression Annotation." *Language Resources and Evaluation (published online)*.
5. A. Alabdullah, L. Han and C. Lin. "Ara-HOPE: Human-Centric Post-Editing Evaluation for Dialectal Arabic to Modern Standard Arabic Translation." *VarDial at EACL 2026*.
6. S. Gladkoff et al.. "Looking under the Wrong Lamppost: On the Limitations of Automated Translation Quality Estimation." *ICNLSP 2026 (accepted)*.
7. L. Bogdanova et al.. "FLANS at SemEval-2026 Task 7: RAG with Open-Sourced Smaller LLMs for Everyday Knowledge Across Diverse Languages and Cultures." *SemEval 2026 (accepted)*.
8. K. Yin Hong et al.. "CANTONMT: Investigating Back-Translation and Model-Switch Mechanisms for Cantonese-English Neural Machine Translation." *ACM TALLIP (accepted)*.
9. L. Han et al.. "Neural Machine Translation of Clinical Text: An Empirical Investigation into Multilingual Pre-Trained Language Models and Transfer-Learning." *Frontiers in Digital Health 6 (2024)*.
10. S. Gladkoff and L. Han. "HOPE: A Task-Oriented and Human-Centric Evaluation Framework Using Professional Post-Editing Towards More Effective MT Evaluation." *LREC 2022*.
11. L. Han et al.. "cushLEPOR: Customising hLEPOR Metric Using Optuna for Higher Agreement with Human Judgments or Pre-trained Language Model LaBSE." *WMT 2021*.
12. L. Han et al.. "Chinese Character Decomposition for Neural MT with Multi-Word Expressions." *NoDaLiDa 2021*.
13. L. Han, G. Jones and A. Smeaton. "MultiMWE: Building a Multi-lingual Multi-Word Expression (MWE) Parallel Corpora." *LREC 2020*.
14. L. Han, G. Jones and A. Smeaton. "AlphaMWE: Construction of Multi-lingual Parallel Corpora with MWE Annotations." *MWE-LEX at COLING 2020*.
15. L. Han, D. F. Wong and L. S. Chao. "LEPOR: A Robust Evaluation Metric for Machine Translation with Augmented Factors." *COLING 2012*.

## Complete Peer-Reviewed Publications and Accepted Papers

*Complete list transcribed from the peer-reviewed section of the November 2026 CV. Author lists are abbreviated for space; publication and acceptance statuses follow the source CV.*

### 2026

1. L. Han et al.. "Mind the Gap: Exposing LLM Translation Blind Spots Using the AlphaMWE Multilingual Parallel Corpus." *WMT 2026*.
2. J. Liang and L. Han. "MetaHOPE: A Metaphor-Oriented Evaluation Framework for Analysing MT and LLM Translation Errors." *ICNLSP 2026*.

3. J. Liang and L. Han. “Towards End-to-End Multilingual Metaphor Processing: Integrating Detection, Translation, and Evaluation." *AACL-IJCNLP 2026 Student Research Workshop (accepted).*
4. M. Zhyrko, S. Verberne and L. Han. “Time as Structure: Temporal Dependency Graphs for Verifiable Deadline Computation over Legal Documents." *AACL-IJCNLP 2026 Student Research Workshop (accepted).*
5. M. C. Staiano et al.. “SIREN: Situation-aware Index for Response and Emergency ActioNability in Crisis Translation." *LT4CPR at AACL-IJCNLP 2026 (accepted).*
6. S. Gladkoff et al.. “Looking under the Wrong Lamppost: On the Limitations of Automated Translation Quality Estimation." *ICNLSP 2026 (accepted).*
7. M. Zhyrko et al.. “DreamerNLplus: Interpretable Modeling of Mental Health Dynamics from Social Media Timelines using Hybrid Rule-Based and RAG Methods." *CLPsych 2026 (accepted).*
8. A. J. W. de Vink et al.. “QuadAI at SemEval-2026 Task 3: Ensemble Learning of Hybrid RoBERTa and LLMs for Dimensional Aspect-Based Sentiment Analysis." *SemEval 2026 (accepted).*
9. L. Bogdanova et al.. “FLANS at SemEval-2026 Task 7: RAG with Open-Sourced Smaller LLMs for Everyday Knowledge Across Diverse Languages and Cultures." *SemEval 2026 (accepted).*
10. A. Alabdullah, L. Han and C. Lin. “Ara-HOPE: Human-Centric Post-Editing Evaluation for Dialectal Arabic to Modern Standard Arabic Translation." *VarDial at EACL 2026.*
11. T.-C. Ionescu et al.. “Analyzing Cancer Patients’ Experiences with Embedding-based Topic Modeling and LLMs." *CLIN Journal (accepted).*
12. L. Han et al.. “Towards a Resource for Multilingual Lexicons: An MT Assisted and Human-in-the-Loop Multilingual Parallel Corpus with Multi-Word Expression Annotation." *Language Resources and Evaluation (published online).*

## 2025

1. L. Ren, Y. M. Ng and L. Han. “MaLei at MultiClinSUM: Summarisation of Clinical Documents using Perspective-Aware Iterative Self-Prompting with LLMs." *BioASQ Workshop at CLEF 2025.*
2. A. J. W. de Vink, N. Amat-Lefort and L. Han. “ReviewGraph: A Knowledge Graph Embedding Based Framework for Review Rating Prediction with Sentiment Features." *ICKG 2025 (listed as forthcoming in source CV).*
3. M. C. Staiano et al.. “ITALERT: Assessing the Quality of LLMs and NMT in Translating Italian Emergency Response Text." *MT Summit 2025.*
4. Z. Ling et al.. “MALEI at the PLABA Track of TAC-2024: RoBERTa for Task 1 - LLaMA3.1 and GPT-4o for Task 2." *TREC Conference (listed under 2025).*
5. S. Belkadi et al.. “MLM4SynMed: Masked Language Modelling for Synthetic Free-text Medical Records Generation." *IEEE ICHI 2025.*
6. D. Shaji et al.. “Advancing De-identification of Clinical Texts using Biomed-Clinical BERTs and Comprehensive Privacy Risk Assessment." *IEEE ICHI 2025 (poster).*
7. L. Ren et al.. “Synthetic4Health: Generating Annotated Synthetic Clinical Letters." *Frontiers in Digital Health 7 (2025).*
8. L. Ren et al.. “Beyond Reconstruction: Generating Privacy-Preserving Clinical Letters." *CL4Health at NAACL 2025.*
9. P. Romero, L. Han and G. Nenadic. “Medication Extraction and Entity Linking using Stacked and Voted Ensembles on LLMs." *CL4Health at NAACL 2025.*
10. P. Romero et al.. “The Manchester Bees at PerAnsSumm 2025: Iterative Self-Prompting with Claude and o1 for Perspective-aware Healthcare Answer Summarisation." *CL4Health at NAACL 2025.*
11. S. Belkadi et al.. “LT3: Generating Medication Prescriptions with Conditional Transformer." *CL4Health at NAACL 2025.*

## 2024

1. K. Yin Hong et al.. "CantonMT: Cantonese-English Neural Machine Translation Looking into Evaluations." *AMTA 2024*.
2. K. Yin Hong et al.. "CANTONMT: Investigating Back-Translation and Model-Switch Mechanisms for Cantonese-English Neural Machine Translation." *ACM TALLIP (accepted)*.
3. A. Lommel et al.. "The Multi-Range Theory of Translation Quality Measurement: MQM Scoring Models and Statistical Quality Control." *AMTA 2024*.
4. L. Han et al.. "Neural Machine Translation of Clinical Text: An Empirical Investigation into Multilingual Pre-Trained Language Models and Transfer-Learning." *Frontiers in Digital Health 6 (2024)*.
5. K. Yin Hong et al.. "CantonMT: Cantonese to English NMT Platform with Fine-Tuned Models Using Synthetic Back-Translation Data." *EAMT 2024*.
6. Z. Li et al.. "Investigating Large Language Models and Control Mechanisms to Improve Text Readability of Biomedical Abstracts." *IEEE ICHI 2024*.
7. S. Gladkoff et al.. "MTUncertainty: Assessing the Need for Post-editing of Machine Translation Outputs by Fine-tuning OpenAI LLMs." *EAMT 2024*.

## 2023

1. Z. Li et al.. "BeeManc at the PLABA Track of TAC 2023: Investigating LLMs and Controllable Attributes for Improving Biomedical Text Readability." *TAC 2023*.
2. S. Belkadi et al.. "Generating Medical Instructions with Conditional Transformer." *SyntheticData4ML at NeurIPS 2023*.
3. S. Belkadi et al.. "Exploring the Value of Pre-trained Language Models for Clinical Named Entity Recognition." *IEEE BigData 2023 workshop*.
4. H. Tu, L. Han and G. Nenadic. "Extraction of Medication and Temporal Relation from Clinical Text using Neural Language Models." *IEEE BigData 2023 workshop*.
5. B. Griciute, L. Han and G. Nenadic. "Topic Modelling of Swedish Newspaper Articles about Coronavirus: A Case Study using Latent Dirichlet Allocation Method." *IEEE ICHI 2023*.
6. N. H. Mohamed et al.. "AlphaMWE-Arabic: Arabic Edition of Multilingual Parallel Corpora with Multiword Expression Annotations." *RANLP 2023*.
7. S. Gladkoff, L. Han and G. Nenadic. "Student's t-Distribution: On Measuring the Inter-Rater Reliability When the Observations are Scarce." *RANLP 2023*.
8. Y. Cui, L. Han and G. Nenadic. "MedTem2.0: Prompt-based Temporal Classification of Treatment Events from Discharge Summaries." *ACL 2023 Student Research Workshop*.
9. S. Schulz et al.. "Towards Principles of Ontology-based Annotation of Clinical Narratives." *ICBO 2023*.
10. L. Han et al.. "Investigating Massive Multilingual Pre-Trained Machine Translation Models for Clinical Domain via Transfer Learning." *ClinicalNLP at ACL 2023*.

## 2022

1. L. Han et al.. "Examining Large Pre-Trained Language Models for Machine Translation: What You Don't Know About It." *WMT 2022*.
2. S. Gladkoff and L. Han. "HOPE: A Task-Oriented and Human-Centric Evaluation Framework Using Professional Post-Editing Towards More Effective MT Evaluation." *LREC 2022*.
3. S. Gladkoff et al.. "Measuring Uncertainty in Translation Quality Evaluation (TQE)." *LREC 2022*.

## 2021

1. L. Han. “Meta-Evaluation of Machine Translation Evaluation Methods." *SIG-MET at ASIS&T 2021 (tutorial paper).*
2. L. Han et al.. “cushLEPOR: Customising hLEPOR Metric Using Optuna for Higher Agreement with Human Judgments or Pre-trained Language Model LaBSE." *WMT 2021.*
3. G. Erofeev et al.. “cushLEPOR Uses LaBSE Distilled Knowledge to Improve Correlation with Human Translation Evaluations." *MT Summit 2021 User Track.*
4. L. Han et al.. “Chinese Character Decomposition for Neural MT with Multi-Word Expressions." *NoDaLiDa 2021.*
5. L. Han, G. Jones and A. Smeaton. “Translation Quality Assessment: A Brief Survey on Manual and Automatic Methods." *MoTra 2021.*

## 2020

1. L. Han, G. Jones and A. Smeaton. “MultiMWE: Building a Multi-lingual Multi-Word Expression (MWE) Parallel Corpora." *LREC 2020.*
2. L. Han, G. Jones and A. Smeaton. “AlphaMWE: Construction of Multi-lingual Parallel Corpora with MWE Annotations." *MWE-LEX at COLING 2020.*

## 2018

1. L. Han and S. Kuang. “Incorporating Chinese Radicals into Neural Machine Translation: Deeper than Character Level." *ESSLLI 2018.*
2. E. Moreau et al.. “Semantic Re-Ranking of CRF Label Sequences for Verbal Multi-Word Expression Extraction." *Language Science Press, edited volume (2018).*

## 2017

1. A. Maldonado et al.. “Detection of Verbal Multi-Word Expressions via Conditional Random Fields with Syntactic Dependency Features and Semantic Re-Ranking." *MWE Workshop at EACL 2017.*

## 2015

1. L. Han et al.. “Chinese Named Entity Recognition with Graph-based Semi-supervised Learning Model." *SIGHAN at ACL-IJCNLP 2015.*

## 2014

1. L. Han et al.. “A Universal Phrase Tagset for Multilingual Treebanks." *CCL and NLP-NABD 2014, Springer LNAI 8801.*
2. L. Han et al.. “Unsupervised Quality Estimation Model for English to German Translation and Its Application in Extensive Supervised Evaluation." *The Scientific World Journal (2014).*

## 2013

1. L. Han et al.. “Language-independent Model for Machine Translation Evaluation with Reinforced Factors." *MT Summit XIV (2013).*
2. L. Han et al.. “A Description of Tunable Machine Translation Evaluation Systems in WMT13 Metrics Task." *WMT 2013.*
3. L. Han et al.. “Quality Estimation for Machine Translation Using the Joint Method of Evaluation Criteria and Statistical Modeling." *WMT 2013.*

4. L. Han et al.. “Phrase Tagset Mapping for French and English Treebanks and Its Application in Machine Translation Evaluation." *GSCL 2013, Springer LNCS 8105*.
5. L. Han et al.. “A Study of Chinese Word Segmentation Based on the Characteristics of Chinese." *GSCL 2013, Springer LNCS 8105*.
6. L. Han et al.. “Automatic Machine Translation Evaluation with Part-of-Speech Information." *TSD 2013, Springer LNAI 8082*.
7. L. Han et al.. “Chinese Named Entity Recognition with Conditional Random Fields in the Light of Chinese Characteristics." *IIS 2013, Springer LNCS 7912*.

### 2012

1. L. Han, D. F. Wong and L. S. Chao. “LEPOR: A Robust Evaluation Metric for Machine Translation with Augmented Factors." *COLING 2012*.

## Invited Talks and Tutorials

**Invited Talks**

- **University of Stuttgart, IMS/NLP Institute** (in person; travel funded), Stuttgart, Germany, 11 October 2022. “Cracking Multi-Word Expressions in a Nutshell: Neural MT Walks on Thin Ice When Facing MWEs.”
- **Lancaster University NLP Group** (online), 5 March 2025. “Clinical NLP and the 4D PICTURE Project.” Recording.

**Conference Tutorials**

- **LREC 2022**, Marseille, France, 20 June 2022. “Meta-Evaluation of Translation Evaluation Methods: A Systematic Up-to-Date Overview.”
- **LREC 2026**, Palma, Spain, 12 May 2026. Half-day main-conference tutorial: “HealthcareNLP: Where Are We and What’s Next?”

## Manuscript Under Review / Preprint

J. Liang, T. Duan, A. Zhang, Y. Huang, X. Deng, and **L. Han**. “Towards a Resource for News Domain Bilingual Metaphor-Annotated Corpus: LLMs Walk on Thin Ice When Translating Metaphors." Submitted to *Language Resources and Evaluation*; preprint via Leiden University Scholarly Publications. [Research supervisor]

## Languages and Research Profile

Chinese (native); English (professional working proficiency); basic/learning: Dutch, German, Italian, Polish and Spanish.
Google Scholar citations: 1,300+; h-index 21; i10-index 36 (CV figures as of October 2026).
ORCID: 0000-0002-3221-2185  Google Scholar publication list